

Gayatri Vidya Parishad Institute of Health Care and Medical Technology



Subject wise COMPETENCIES

Phase I MBBS

Competency Based Medical Education

The Indian Medical Graduate · CBME Curriculum 2024

Goals · Topics · Competencies

Anatomy · Physiology · Biochemistry · Community Medicine

Department of Medical Education

Gayatri Vidya Parishad Institute of Health Care and Medical Technology, Visakhapatnam

Academic Year 2026–27

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HOW TO USE THIS HANDBOOK

This handbook is for **MBBS Phase I students of the academic year 2026–27** of Gayatri Vidya Parishad Institute of Health Care and Medical Technology, Visakhapatnam. It is a practical companion for class, self-study and family adoption postings. It is **not a textbook**. Use it to know what you are expected to learn, demonstrate and perform.

What you will find here

- A short explanation of CBME, the five roles of the Indian Medical Graduate (IMG), and how Phase I is organised.
- Subject goals for Anatomy, Physiology, Biochemistry and Community Medicine.
- Topics and numbered competencies for these Phase I subjects, as given in the NMC CBME 2024 guidelines.

How to read a competency code

Each competency has a code such as **AN1.1**, **PY2.3**, **BC3.1** or **CM1.4**.

- **Letters** show the subject: AN Anatomy, PY Physiology, BC Biochemistry, CM Community Medicine.

- The **first number** is the topic.

- The **second number** is the competency within that topic.

Example: **AN10.3** is Anatomy, Topic 10 (Axilla, Shoulder and Scapular region), third competency.

How to study with this handbook

- Read the subject goals first. They tell you the purpose of the whole subject.
- Before a class or practical, look up the topic and tick the competencies that will be covered.
- After teaching, check whether you can describe, demonstrate or perform what the competency asks.
- Community Medicine teaching begins in Phase I (including the Family Adoption Programme) and continues in later phases. This handbook lists the CM competencies so that you can start using them from the first year.

Source: National Medical Commission, Competency Based Medical Education (CBME) Curriculum 2024. Prepared by the Department of Medical Education.

WHAT IS CBME?

Competency based learning means designing and implementing the medical education curriculum so that it focuses on the **desired and observable ability in real-life situations**. In simple words: it is not enough to recall facts in an examination. By the end of training you should be able to **do** what a doctor is expected to do, with the right knowledge, skill and attitude.

This handbook follows the National Medical Commission (NMC) **Competency Based Medical Education (CBME) Curriculum 2024**. These guidelines apply from the MBBS batch of 2024–25 onwards, including your batch of 2026–27.

THE INDIAN MEDICAL GRADUATE (IMG)

The Indian Medical Graduate is the doctor you are being trained to become. In order to fulfil this role, the IMG would have obtained the following set of competencies at the time of graduation. Keep these five roles in mind throughout Phase I. Your subject competencies are the first steps towards them.

Clinician

who understands and provides preventive, promotive, curative, palliative and holistic care with compassion.

- Demonstrate knowledge of normal and abnormal human structure, function and development from molecular, cellular, biologic, clinical, behavioural and social perspectives.
- Demonstrate knowledge of medico-legal, societal, ethical and humanitarian principles that influence health care, and of national health policies including the National Health Mission.
- Elicit and record a complete, relevant history and perform a physical examination that is contextual to gender, age, vulnerability, beliefs and values.
- Demonstrate effective clinical problem solving, choose appropriate diagnostic tests, prescribe rational therapy, maintain records, and identify patients who require referral.

Leader and member of the health care team and system

- Work effectively with colleagues in an inter-professional team, respecting diversity of roles and competencies.
- Function as a health-care team leader in primary and secondary settings and utilise the health system in a cost-effective, fair manner aligned with national priorities.
- Participate in quality improvement, patient safety, health promotion and disease prevention, including lifestyle diseases and cancer.

Communicator

with patients, families, colleagues and community.

- Communicate adequately, sensitively, effectively and respectfully in a language that patients understand.
- Establish professional relationships that are humane, ethical, empathetic and trustworthy, and that respect confidentiality, privacy and shared decision-making.

Lifelong learner

committed to continuous improvement of skills and knowledge.

- Perform objective self-assessment, continue learning, refine existing skills and acquire new skills, and apply them to patient care.
- Search and critically evaluate the medical literature and identify an appropriate career pathway.

Professional

who is committed to excellence, is ethical, responsive and accountable to patients, the profession and the community.

- Practice selflessness, integrity, responsibility, accountability and respect, and maintain professional boundaries.
- Recognise and manage ethical and professional conflicts, and abide by prescribed ethical and legal codes of conduct.

PHASE I AT A GLANCE

Phase I lasts **12 months**, including a Foundation Course of two weeks and the university examination. About 39 weeks (1521 hours) are available for teaching–learning. The rest of the year is for examinations, vacation and public holidays.

What Phase I includes

- Anatomy, Physiology and Biochemistry.
- Introduction to Community Medicine, Humanities, and the Attitude, Ethics & Communication (AETCOM) module.
- Family Adoption Programme (FAP) through village outreach: each student shall adopt a minimum of three families, and preferably five.
- Early clinical exposure, simulation-based learning, alignment and integration, and the pandemic module (integrated with subjects).

Foundation Course (first two weeks, 80 hours)

- Orientation, including History of Indian Medicine (15 hours).
- Skills module (15 hours).
- Community orientation (5 hours).
- Professional development and ethics, including mental health (20 hours).
- Language and computer skills, including clinico-laboratory communication (10 hours).
- Sports and extra-curricular activities (15 hours).

Teaching hours in Phase I

Subject	LGT	SGT*	SDL	Total
Foundation Course	—	—	—	80
Anatomy	180	430	10	620

Physiology	130	305	10	445
Biochemistry	82	157	10	249
Early Clinical Exposure	—	27	—	27
Community Medicine	20	20	—	40
Family Adoption Programme	—	24	—	24
AETCOM	—	26	—	26
Sports / extra-curricular	—	—	—	10
Total	412	989	30	1521

*SGT = small-group teaching / practicals / tutorials / seminars. LGT = large-group teaching. SDL = self-directed learning. ECE hours are divided equally among Anatomy, Physiology and Biochemistry. AETCOM is a longitudinal programme.

Family Adoption Programme — Phase I targets

- Build rapport and connect with the adopted families; learn communication and inspire trust.
- Understand the community set-up of the region and mobilise families for screening programmes.
- Undertake a detailed family study, prepare a family diagnosis (disease, ill-health, malnutrition, risk factors, scope for health promotion), and formulate objectives for each family.

Eligibility for the Phase I university examination (keep these in mind from day one)

- **Attendance:** at least 75% in each subject.
- **Internal assessment:** at least 50% combined in theory and practical, with a minimum of 40% in each; at least three theory and practical internal

assessments, including the pre-university examination.

- **Log book:** required certifiable competencies for the phase must be completed and recorded.

Internal assessment is not added to the university (summative) marks, but you must meet these criteria to be allowed to appear. Community Medicine internal assessment continues across phases (25 marks in Phase I towards the final 100).

ANATOMY

82 topics · 413 competencies · Phase I MBBS

Subject Goals

At the end of anatomy teaching, a student should be able to demonstrate:

- i. Comprehension of normal structure, development and genetic pattern of organ and organ systems, as well as the clinical correlation of structures involved in diseases and its anatomical basis.
- ii. Comprehension of the normal disposition, clinically relevant inter-relationships, functional and cross-sectional Anatomy of the various organs and structures of the body.
- iii. Identification of the microscopic structure of various organs and tissues with the functions, as a prerequisite for understanding the altered state in various disease processes.
- iv. Basic principles and sequential development of the organs and systems; recognise the critical stages of development and the effects of common teratogens, genetic mutations and environmental hazards.
- v. Principles of karyotyping and identify the gross congenital anomalies.
- vi. Principles of newer imaging techniques and interpretation of CT scan, sonogram, MRI and Angiography.

Topics and Competencies

Topic 1: Anatomical terminology

AN1.1	Describe & Demonstrate normal anatomical position, various planes, relation, comparison, laterality & movements in the human body
AN1.2	Describe composition of bone and bone marrow

Topic 2: General features of bones & Joints

AN2.1	Describe parts, types, peculiarities of each type, blood and nerve supply of bones.
AN2.2	Describe the laws of ossification, epiphysis, its various types and their importance
AN2.3	Describe special features of a sesamoid bone
AN2.4	Describe various types of cartilage with its structure & distribution in body
AN2.5	Describe & demonstrate various joints with possible movements, subtypes and examples
AN2.6	Explain the concept of nerve supply of joints & Hilton's law

Topic 3: General features of Muscle

AN3.1	Classify & describe muscle tissue according to structure, size, shape, region & action
AN3.2	Describe parts of skeletal muscle and differentiate between tendons and aponeuroses with examples
AN3.3	Explain Shunt and spurt muscles with examples and role in joint movement

Topic 4: General features of skin and fascia

AN4.1	Describe different types of skin & dermatomes in body
AN4.2	Describe & demonstrate structure of skin with its appendages along with clinical anatomy

AN4.3 Describe structure, contents and identify modifications of superficial fascia along with fat distribution in body

AN4.4 Describe & demonstrate modifications of deep fascia with its location, function & examples

AN4.5 Explain principles of skin incisions and their surgical importance

Topic 5: General features of the cardiovascular system

AN5.1 Differentiate between blood vascular and lymphatic system

AN5.2 Differentiate between pulmonary and systemic circulation

AN5.3 Describe general differences between arteries, veins and sinuses

AN5.4 Explain functional and gross structural differences between elastic, muscular arteries and arterioles

AN5.5 Describe portal system giving examples

AN5.6 Describe the concept of anastomoses and collateral circulation, its different sites & significance of end arteries

AN5.7 Explain function of meta-arterioles, precapillary sphincters, arterio-venous anastomoses

AN5.8 Describe thrombosis, infarction & aneurysm

Topic 6: General Features of lymphatic system

AN6.1 Describe the components and functions of the lymphatic system

AN6.2 Describe structure of lymph capillaries & mechanism of lymph circulation

AN6.3 Explain the concept of lymphoedema and spread of tumors via lymphatics and venous system

Topic 7: Introduction to the nervous system

AN7.1 Describe general plan of nervous system with components of central, peripheral & autonomic nervous systems

AN7.2 List components of nervous tissue and their functions

AN7.3 Describe parts of a neuron and classify them based on number of neurites, size & function

AN7.4 Describe structure of a typical spinal nerve

AN7.5 Describe principles of sensory and motor innervation of muscles

AN7.6 Describe concept of loss of innervation of a muscle with its applied anatomy

AN7.7 Describe various types of synapse

AN7.8 Describe differences between sympathetic and spinal ganglia

Topic 8: Features of individual bones (Upper Limb)

AN8.1 Identify the given bone, its side, anatomical position, joint formation, important features and clinical anatomy (clavicle, scapula, humerus, radius, ulna, carpal bones)

AN8.2 Demonstrate important muscle attachments on the given bone

AN8.3 Identify and name various bones in articulated hand, Specify the parts of metacarpals and phalanges and enumerate the peculiarities of pisiform

AN8.4 Describe scaphoid fracture and explain the anatomical basis of avascular necrosis

Topic 9: Pectoral region

AN9.1 Describe attachment, nerve supply & action of pectoralis major and pectoralis minor and describe clavipectoral fascia

AN9.2 Describe the location, extent, deep relations, structure, blood supply, lymphatic drainage, microanatomy and applied anatomy of breast

AN9.3 Describe development of breast, associated age changes and congenital anomalies

Topic 10: Axilla, Shoulder and Scapular region

AN10.1 Identify & describe boundaries and contents of axilla

AN10.2 Identify, describe and demonstrate the origin, extent, course, parts, relations and branches of axillary artery & tributaries of axillary vein

AN10.3 Describe, identify and demonstrate formation, branches, relations, area of supply of branches, course and relations of terminal branches of brachial plexus

AN10.4 Describe the anatomical groups of axillary lymph nodes and specify their areas of drainage

AN10.5 Explain variations in formation of brachial plexus

AN10.6 Explain the anatomical basis of clinical features of Erb's palsy and Klumpke's paralysis

AN10.7 Describe axillary lymph nodes, areas of drainage and anatomical basis of their enlargement

AN10.8 Describe, identify and demonstrate the position, attachment, nerve supply and actions of trapezius and latissimus dorsi

AN10.9 Describe the arterial anastomosis around the scapula and mention the boundaries of triangle of auscultation

AN10.10 Describe and identify the deltoid and rotator cuff muscles along with their nerve supply and clinical anatomy

AN10.11 Describe & demonstrate attachment, action and clinical anatomy of serratus anterior muscle

AN10.12 Describe and demonstrate shoulder joint for– type, articular surfaces, capsule, synovial membrane, ligaments, relations, movements, muscles involved, blood supply, nerve supply and applied anatomy

AN10.13 Explain anatomical basis of Injury to axillary nerve during intramuscular injections

Topic 11: Arm & Cubital fossa

AN11.1 Describe and demonstrate muscle groups of upper arm with emphasis on biceps and triceps brachii

AN11.2 Identify & describe origin, course, relations, branches (or tributaries), termination of important nerves and vessels in arm

AN11.3 Describe the anatomical basis of Venipuncture of cubital veins

AN11.4 Describe the anatomical basis of Saturday night paralysis

AN11.5 Identify & describe boundaries and contents of cubital fossa

AN11.6 Describe the anastomosis around the elbow joint

Topic 12: Forearm & hand

AN12.1	Describe and demonstrate important muscle groups of ventral forearm with attachments, nerve supply and actions
AN12.2	Identify & describe origin, course, relations, branches (or tributaries), termination of important nerves and vessels of forearm
AN12.3	Identify & describe flexor retinaculum with its attachments
AN12.4	Explain anatomical basis of carpal tunnel syndrome
AN12.5	Identify & describe small muscles of hand. Also describe movements of thumb and muscles involved
AN12.6	Describe & demonstrate movements of thumb and muscles involved
AN12.7	Identify & describe course and branches of important blood vessels and nerves in hand
AN12.8	Describe anatomical basis of Claw hand
AN12.9	Identify & describe fibrous flexor sheaths, ulnar bursa, radial bursa and digital synovial sheaths
AN12.10	Explain infection of fascial spaces of palm
AN12.11	Identify, describe and demonstrate important muscle groups of dorsal forearm with attachments, nerve supply and actions
AN12.12	Identify & describe origin, course, relations, branches (or tributaries), termination of important nerves and vessels of back of forearm
AN12.13	Describe the anatomical basis of Wrist drop

AN12.14 Identify & describe compartments deep to extensor retinaculum and describe the boundaries and contents of anatomical snuff box.

AN12.15 Identify & describe extensor expansion formation

Topic 13: General Features, Joints, radiographs & surface marking

AN13.1	Describe and explain Fascia of upper limb and compartments, veins of upper limb and its lymphatic drainage
AN13.2	Describe dermatomes of upper limb
AN13.3	Identify & describe the type, articular surfaces, capsule, synovial membrane, ligaments, relations, movements, blood and nerve supply of elbow joint, proximal and distal radio-ulnar joints, wrist joint & first carpometacarpal joint
AN13.4	Describe Sternoclavicular joint, Acromioclavicular joint, Carpometacarpal joints & Metacarpophalangeal joint
AN13.5	Identify the bones and joints of upper limb seen in anteroposterior and lateral view radiographs of shoulder region, arm, elbow, forearm and hand
AN13.6	Identify & demonstrate important bony landmarks of upper limb: Jugular notch, sternal angle, acromial angle, spine of the scapula, vertebral level of the medial end and Inferior angle of the scapula
AN13.7	Identify & demonstrate surface projection of: Cephalic and basilic vein, Palpation of Brachial artery, Radial artery, Testing of muscles: Trapezius, pectoralis major, serratus anterior, latissimus dorsi, deltoid, biceps brachii, Brachioradialis

AN13.8 Describe development of upper limb

Topic 14: Features of individual bones (Lower Limb)

AN14.1 Identify the given bone, its side, anatomical position, joint formation, important features and clinical anatomy (hip bone, femur, tibia fibula, tarsal bones)

AN14.2 Identify & describe joints formed by the given bone

AN14.3 Describe the importance of ossification of lower end of femur & upper end of tibia, and explain violation of law of ossification in fibula

AN14.4 Identify and name various bones in the articulated foot with individual muscle attachment

Topic 15: Front & Medial side of thigh

AN15.1 Describe and demonstrate origin, course, relations, branches (or tributaries), termination of important nerves and vessels of anterior thigh

AN15.2 Describe and demonstrate major muscles with their attachment, nerve supply and actions

AN15.3 Describe and demonstrate boundaries, floor, roof and contents of femoral triangle

AN15.4 Explain anatomical basis of Psoas abscess & Femoral hernia

AN15.5 Describe and demonstrate adductor canal with its contents

Topic 16: Gluteal region & back of thigh

AN16.1 Describe and demonstrate major muscles with their attachment, nerve supply and actions.

AN16.2 Describe and demonstrate structures under the cover of gluteus maximus. Also explain the anatomical basis of sciatic nerve injury during gluteal intramuscular injections

AN16.3 Explain the anatomical basis of Trendelenburg sign

AN16.4 Describe and demonstrate the hamstrings group of muscles with their attachment, nerve supply and actions

AN16.5 Describe and demonstrate the origin, course, relations, branches (or tributaries), termination of important nerves and vessels on the back of thigh

AN16.6 Describe and demonstrate the boundaries, roof, floor, contents and relations of popliteal fossa with its clinical anatomy

Topic 17: Hip Joint

AN17.1 Describe and demonstrate the type, articular surfaces, capsule, synovial membrane, ligaments, relations, movements and muscles involved, blood and nerve supply, bursae around the hip joint

AN17.2 Describe anatomical basis of complications of fracture neck of femur

AN17.3 Describe dislocation of hip joint and surgical hip replacement

Topic 18: Knee joint, Anterior compartment of leg & dorsum of foot

AN18.1 Describe and demonstrate major muscles of anterior compartment of leg with their attachment, nerve supply and actions

AN18.2 Describe and demonstrate origin, course, relations, branches (or tributaries), termination of important nerves and vessels of anterior compartment of leg

AN18.3 Explain the anatomical basis of foot drop

AN18.4 Describe and demonstrate the type, articular surfaces, capsule, synovial membrane, ligaments, relations, movements and muscles involved, nerve supply, bursae around the knee joint along with anastomosis around the knee joint

AN18.5 Explain the anatomical basis of locking and unlocking of the knee joint

AN18.6 Describe knee joint injuries with its applied anatomy

AN18.7 Explain anatomical basis of Osteoarthritis

Topic 19: Back of Leg & Sole

AN19.1 Describe and demonstrate the major muscles of back of leg with their attachment, nerve supply and actions

AN19.2 Describe and demonstrate the origin, course, relations, branches (or tributaries), termination of important nerves and vessels of back of leg

AN19.3 Explain the concept of “Peripheral heart”

AN19.4 Explain the anatomical basis of rupture of calcaneal tendon

AN19.5 Describe factors maintaining importance arches of the foot with its importance

AN19.6 Explain the anatomical basis of Flat foot & Club foot

AN19.7 Explain the anatomical basis of Metatarsalgia & Plantar fasciitis

Topic 20: General Features, Joints, radiographs & surface marking

AN20.1 Describe and demonstrate the type, articular surfaces, capsule, synovial membrane, ligaments, relations, movements and muscles involved, blood and nerve supply of tibiofibular and ankle joint

AN20.2 Describe the subtalar and transverse tarsal joints

AN20.3 Describe and demonstrate Fascia lata, Venous drainage, Lymphatic drainage, Retinacula & Dermatomes of lower limb

AN20.4 Explain anatomical basis of enlarged inguinal lymph nodes

AN20.5 Explain anatomical basis of varicose veins and deep vein thrombosis

AN20.6 Identify the bones and joints of lower limb seen in anteroposterior and lateral view radiographs of various regions of lower limb

AN20.7 Identify & demonstrate important bony landmarks of lower limb: - Vertebral levels of highest point of iliac crest, posterior superior iliac spines, iliac tubercle, pubic tubercle, ischial tuberosity, adductor tubercle, -Tibial tuberosity, head of fibula, -Medial and lateral malleoli, Condyles of femur and tibia, sustentaculum tali, tuberosity of fifth metatarsal, tuberosity of the navicular

AN20.8 Identify & demonstrate palpation of femoral, popliteal, posterior tibial, anterior tibial & dorsalis pedis arteries in a simulated environment

AN20.9 Demonstrate surface projection of: femoral, popliteal, dorsalis pedis, post tibial arteries, Mid inguinal point, femoral nerve, Saphenous opening, Sciatic, tibial, common peroneal & deep peroneal nerve, Great and small saphenous veins

AN20.10 Describe basic concept of development of lower limb

Topic 21: Thoracic cage

AN21.1 Identify and describe the salient features of sternum, typical rib and typical thoracic vertebra.

AN21.2 Identify & describe the features of atypical ribs and atypical thoracic vertebrae.

AN21.3 Describe & demonstrate the boundaries of thoracic inlet, cavity and outlet along with its applied aspect.(Thoracic inlet Syndrome)

AN21.4 Describe & demonstrate extent, attachments, direction of fibres, nerve supply and actions of intercostal muscles

AN21.5 Describe & demonstrate origin, course, relations and branches of a typical intercostal nerve

AN21.6 Mention origin, course and branches/ tributaries of: anterior & posterior intercostal vessels internal thoracic vessels

AN21.7 Mention the origin, course, relations and branches of atypical intercostal nerve superior intercostal artery, subcostal artery

AN21.8 Describe & demonstrate type, articular surfaces & movements of manubriosternal, costovertebral, costotransverse and xiphisternal joints

AN21.9 Describe & demonstrate mechanics and types of respiration

AN21.10 Describe costochondral and interchondral joints

AN21.11 Mention boundaries and contents of the superior, anterior, middle and posterior mediastinum

Topic 22: Heart & Pericardium

AN22.1 Describe & demonstrate subdivisions, sinuses in pericardium, blood supply and nerve supply of pericardium

AN22.2 Describe & demonstrate external and internal features of each chamber of heart

AN22.3 Describe & demonstrate origin, course and branches of coronary arteries

AN22.4 Describe anatomical basis of ischaemic heart disease

AN22.5 Describe & demonstrate the formation, course, tributaries and termination of coronary sinus

AN22.6 Describe the fibrous skeleton of heart

AN22.7 Mention the parts, position and arterial supply of the conducting system of heart

Topic 23: Mediastinum

AN23.1 Describe & demonstrate the external appearance, relations, blood supply, nerve supply, lymphatic drainage and applied anatomy of oesophagus

AN23.2 Describe & demonstrate the extent, relations and tributaries of thoracic duct and enumerate its applied anatomy.

AN23.3	Describe & demonstrate origin, course, relations, tributaries and termination of superior vena cava, azygos, hemiazygos and accessory hemiazygos veins
AN23.4	Mention the extent, branches and relations of arch of aorta & descending thoracic aorta
AN23.5	Identify & Mention the location and extent of thoracic sympathetic chain
AN23.6	Describe the splanchnic nerves

Topic 24: Lungs & Trachea

AN24.1	Mention the blood supply, lymphatic drainage and nerve supply of pleura, extent of pleura and describe the pleural recesses and their applied anatomy
AN24.2	Identify side, external features and relations of structures which form root of lung & bronchial tree and their clinical correlate
AN24.3	Describe a bronchopulmonary segment with its clinical anatomy
AN24.4	Identify phrenic nerve & describe its formation & distribution
AN24.5	Mention the blood supply, lymphatic drainage and nerve supply of lungs
AN24.6	Describe the extent, length, relations, blood supply, lymphatic drainage and nerve supply of trachea

Topic 25: Thorax

AN25.1	Identify, draw and label a slide of trachea and lung
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AN25.2	Describe development of pleura, lung & heart
AN25.3	Describe fetal circulation and changes occurring at birth
AN25.4	Describe embryological basis of: 1) atrial septal defect, 2) ventricular septal defect, 3) Fallot's tetralogy & 4) tracheoesophageal fistula
AN25.5	Describe developmental basis of congenital anomalies, transposition of great vessels, dextrocardia, patent ductus arteriosus and coarctation of aorta
AN25.6	Mention development of aortic arch arteries, SVC, IVC and coronary sinus
AN25.7	Identify structures seen on a plain x-ray chest (PA view)
AN25.8	Identify and describe in brief a barium swallow
AN25.9	Demonstrate surface marking of lines of pleural reflection, lung borders and fissures, trachea, heart borders, apex beat & surface projection of valves of heart

Topic 26: Skull osteology

AN26.1	Describe & demonstrate anatomical position of skull, Identify and locate individual skull bones in skull
AN26.2	Describe & demonstrate the features of norma frontalis, verticalis, occipitalis, lateralis and basalis
AN26.3	Describe & demonstrate cranial cavity, its subdivisions, foramina and structures passing through them
AN26.4	Describe & demonstrate morphological features of mandible
AN26.5	Describe & demonstrate features of typical and atypical cervical vertebrae (atlas and axis)

AN26.6	Explain the concept of bones that ossify in membrane
AN26.7	Describe & demonstrate the features of the 7th cervical vertebra

Topic 27: Scalp

AN27.1	Describe & demonstrate the layers of scalp, its blood supply, nerve supply and surgical importance.
AN27.2	Describe emissary veins with its role in the spread of infection from extracranial route to intracranial venous sinuses

Topic 28: Face & parotid region

AN28.1	Describe & demonstrate muscles of facial expression and their nerve supply
AN28.2	Describe sensory innervation of face
AN28.3	Describe & demonstrate origin /formation, course, branches /tributaries of facial vessels
AN28.4	Describe & demonstrate branches of facial nerve with distribution
AN28.5	Describe cervical lymph nodes and lymphatic drainage of head, face and neck
AN28.6	Identify superficial muscles of face, their nerve supply and actions
AN28.7	Explain the anatomical basis of facial nerve palsy
AN28.8	Explain surgical importance of deep facial vein

AN28.9	Describe & demonstrate the parts, borders, surfaces, contents, relations and nerve supply of parotid gland with course of its duct and surgical importance
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AN28.10	Explain the anatomical basis of Frey's syndrome
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Topic 29: Posterior triangle of neck

AN29.1	Describe and demonstrate the boundaries, subdivisions and contents of posterior triangle of neck
AN29.2	Describe & demonstrate attachments, nerve supply, relations and actions of sternocleidomastoid
AN29.3	Explain anatomical basis of Erb's & Klumpke's palsy
AN29.4	Explain anatomical basis of wry neck
AN29.5	Describe & demonstrate attachments of 1) inferior belly of omohyoid, 2)scalenus anterior, 3) scalenus medius & 4) levator scapulae

Topic 30: Cranial cavity

AN30.1	Describe the cranial fossae & identify related structures
AN30.2	Describe & identify major foramina with structures passing through them
AN30.3	Describe & identify dural folds & dural venous sinuses
AN30.4	Describe clinical importance of dural venous sinuses
AN30.5	Explain effect of pituitary tumours on visual pathway

Topic 31: Orbit

AN31.1	Describe & identify extra ocular muscles of eyeball, along with a note on its attachment, action and clinical anatomy
AN31.2	Describe & demonstrate nerves and vessels in the orbit
AN31.3	Describe anatomical basis of Horner's syndrome
AN31.4	Describe the components of lacrimal apparatus
AN31.5	Explain the anatomical basis of oculomotor, trochlear and abducent nerve palsies along with strabismus

Topic 32: Anterior Triangle

AN32.1	Describe boundaries and subdivisions of anterior triangle
AN32.2	Describe & demonstrate boundaries and contents of muscular, carotid, digastric and submental triangles

Topic 33: Temporal and Infratemporal regions

AN33.1	Describe & demonstrate extent, boundaries and contents of temporal and infratemporal fossae
AN33.2	Describe & demonstrate attachments, direction of fibres, nerve supply and actions of muscles of mastication
AN33.3	Describe & demonstrate articulating surface, type & movements of temporomandibular joint
AN33.4	Explain the clinical significance of pterygoid venous plexus
AN33.5	Describe the features of dislocation of temporomandibular joint

Topic 34: Submandibular region

AN34.1	Describe and demonstrate the superficial and deep structures, muscles, nerves, vessels, and glands in the submandibular region
AN34.2	Describe & demonstrate the morphology, relations and nerve supply of submandibular salivary gland & submandibularganglion
AN34.3	Describe the basis of formation of submandibular stones

Topic 35: Deep structures in the neck

AN35.1	Describe the parts, extent, attachments, modifications of deep cervical fascia
AN35.2	Describe & demonstrate location, parts, borders, surfaces, relations, blood supply & applied anatomy of thyroid gland. Also describe the parathyroid glands in brief.
AN35.3	Demonstrate & describe the origin, parts, course & branches subclavian artery
AN35.4	Describe & demonstrate origin, course, relations, tributaries and termination of internal jugular & brachiocephalic veins
AN35.5	Describe and demonstrate extent, drainage & applied anatomy of cervical lymph nodes
AN35.6	Describe and demonstrate the extent, formation, relation & branches of cervical sympathetic chain
AN35.7	Describe the course and branches of IX, X, XI & XII nerve in the neck

AN35.8 Describe the anatomically relevant clinical features of Thyroid swellings

AN35.9 Describe the clinical features of compression of subclavian artery and lower trunk of brachial plexus by cervical rib

AN35.10 Describe the fascial spaces of neck

Topic 36: Mouth, Pharynx & Palate

AN36.1 Describe and demonstrate the structures of the vestibule of the mouth and oral cavity proper.

AN36.2 Describe the 1) morphology, relations, blood supply and applied anatomy of palatine tonsil 2) composition of soft palate

AN36.3 Describe and demonstrate the muscles, nerve supply, blood supply and lymphatic drainage of the pharynx

AN36.4 Describe the components and functions of Waldeyer's lymphatic ring

AN36.5 Describe the pharyngeal spaces. Also describe the boundaries and clinical significance of pyriform fossa

AN36.6 Describe the anatomical basis of tonsillitis, tonsillectomy, adenoids and peri-tonsillar abscess

AN36.7 Describe the clinical significance of Killian's dehiscence

Topic 37: Cavity of Nose

AN37.1 Describe & demonstrate features of nasal septum, lateral wall of nose, their blood supply and nerve supply

AN37.2 Describe location and functional anatomy of paranasal sinuses

AN37.3 Describe anatomical basis of sinusitis & maxillary sinus tumours

Topic 38: Larynx

AN38.1 Describe & demonstrate the morphology, identify structure of the wall, nerve supply, blood supply and actions of intrinsic and extrinsic muscles of the larynx

AN38.2 Describe the anatomical aspects of laryngitis

AN38.3 Describe anatomical basis of recurrent laryngeal nerve injury

Topic 39: Tongue

AN39.1 Describe & demonstrate the morphology, nerve supply, embryological basis of nerve supply, blood supply, lymphatic drainage and actions of extrinsic and intrinsic muscles of tongue

AN39.2 Explain the anatomical basis of hypoglossal nerve palsy

Topic 40: Organs of hearing and equilibrium

AN40.1 Describe & identify the parts, blood supply and nerve supply of external ear

AN40.2 Describe & demonstrate the boundaries, contents, relations and functional anatomy of middle ear and auditory tube

AN40.3 Describe the features of internal ear

AN40.4 Explain anatomical basis of otitis externa and otitis media

AN40.5 Explain anatomical basis of myringotomy

Topic 41: Eyeball

AN41.1 Describe & demonstrate parts and layers of eyeball

AN41.2 Describe the anatomical aspects of cataract, glaucoma & central retinal artery occlusion

AN41.3 Describe the position, nerve supply and actions of intraocular muscles

Topic 42: Back region

AN42.1 Describe and demonstrate the contents of the vertebral canal

AN42.2 Describe & demonstrate the boundaries and contents of Suboccipital triangle

AN42.3 Describe the position, direction of fibres, relations, nerve supply, actions of semispinalis capitis and splenius capitis

Topic 43: Head & neck Joints, Histology, Development, Radiography & Surface marking

AN43.1 Describe & demonstrate the movements with muscles producing the movements of atlantooccipital joint & atlantoaxial joint

AN43.2 Identify, describe and draw the microanatomy of pituitary gland, thyroid, parathyroid gland, tongue, salivary glands, tonsil, epiglottis, cornea, retina

AN43.3 Identify, describe and draw microanatomy of olfactory epithelium, eyelid, lip, sclero-corneal junction, optic nerve, cochlea- organ of corti, pineal gland

AN43.4 Describe the development and developmental basis of congenital anomalies of face, palate, tongue, branchial apparatus, pituitary gland, thyroid gland & eye

AN43.5 Demonstrate- 1) Testing of muscles of facial expression, extraocular muscles, muscles of mastication, 2) Palpation of carotid arteries, facial artery, superficial temporal artery, 3) Location of internal and external jugular veins, 4) Location of hyoid bone, thyroid cartilage and cricoid cartilage with their vertebral levels

AN43.6 Demonstrate surface projection of- Thyroid gland, Parotid gland and duct, Pterion, Common carotid artery, Internal jugular vein, Subclavian vein, External jugular vein, Facial artery in the face & accessory nerve

AN43.7 Identify the anatomical structures in 1) Plain x-ray skull, 2) AP view and lateral view 3) Plain x-ray cervical spine-AP and lateral view 4) Plain x- ray of paranasal sinuses

AN43.8 Describe the anatomical route used for carotid angiogram and vertebral angiogram

AN43.9 Identify anatomical structures in carotid angiogram and vertebral angiogram

Topic 44: Anterior abdominal wall

AN44.1 Describe & demonstrate the Planes (transpyloric, transtubercular, subcostal, lateral vertical, linea alba, linea semilunaris), regions & Quadrants of abdomen

AN44.2 Describe & identify the Fascia, nerves & blood vessels of anterior abdominal wall

AN44.3 Describe the formation of rectus sheath and its contents

AN44.4	Describe & demonstrate extent, boundaries, contents of Inguinal canal including Hesselbach's triangle.
AN44.5	Explain the anatomical basis of inguinal hernia.
AN44.6	Describe & demonstrate attachments of muscles of anterior abdominal wall
AN44.7	Describe common abdominal incisions with example and their clinical importance

Topic 45: Posterior abdominal wall

AN45.1	Describe Thoracolumbar fascia, its different layers, their attachments and extents
AN45.2	Describe & demonstrate Lumbar plexus, its root value, formation, branches and clinical anatomy (compression/injury to the rootlets of lumbar plexus)
AN45.3	Describe and demonstrate back muscles, nerve supply and action

Topic 46: Male external genitalia

AN46.1	Describe & demonstrate coverings, internal structure, side determination, blood supply, nerve supply, lymphatic drainage & descent of testis with its applied anatomy
AN46.2	Describe parts of Epididymis
AN46.3	Describe Penis under following headings: (parts, components, blood supply and lymphatic drainage)
AN46.4	Explain the anatomical basis of Varicocele
AN46.5	Explain the anatomical basis of Phimosis & Circumcision

Topic 47: Abdominal cavity

AN47.1	Describe & demonstrate horizontal and vertical tracing of peritoneum. Also describe boundaries and recesses of Lesser & Greater sac.
AN47.2	Name & identify various peritoneal folds & pouches with its explanation .
AN47.3	Explain anatomical basis of Ascites & Peritonitis
AN47.4	Explain anatomical basis of Subphrenic abscess
AN47.5	Describe & demonstrate major viscera of abdomen under following headings (anatomical position, external and internal features, important peritoneal and other relations, blood supply, nerve supply, lymphatic drainage and applied aspects)
AN47.6	Explain the anatomical basis of Splenic notch, Accessory spleens, Kehr's sign, Different types of vagotomy, Liver biopsy (site of needle puncture), Referred pain in cholecystitis, Obstructive jaundice, Referred pain around umbilicus, Radiating pain of kidney to groin & Lymphatic spread in carcinoma stomach
AN47.7	Demonstrate boundaries of Calot's triangle and mention its clinical importance
AN47.8	Describe & identify the formation, course relations and tributaries of Portal vein, Inferior vena cava & Renal vein
AN47.9	Describe & identify the origin, course, important relations and branches of Abdominal aorta, Coeliac trunk, Superior mesenteric, Inferior mesenteric & Common iliac artery

AN47.10	Describe sites of portosystemic anastomosis, describe its applied anatomy and anatomical correlations
AN47.11	Explain the anatomic basis of hematemesis & caput medusae in portal hypertension
AN47.12	Describe important nerve plexuses of posterior abdominal wall
AN47.13	Describe & demonstrate the attachments, openings, nerve supply & action of the thoracoabdominal diaphragm
AN47.14	Describe the abnormal openings of thoracoabdominal diaphragm and diaphragmatic hernia

Topic 48: Pelvic wall and viscera

AN48.1	Describe & demonstrate the position, features, important peritoneal and other relations, blood supply, nerve supply, lymphatic drainage and clinical aspects of important male & female pelvic viscera.
AN48.2	Describe & identify the muscles of Pelvic diaphragm.
AN48.3	Describe & demonstrate the origin, course, important relations and branches of internal iliac artery
AN48.4	Describe the branches of sacral plexus
AN48.5	Explain the anatomical basis of suprapubic cystostomy, Urinary obstruction in benign prostatic hypertrophy, Retroverted uterus, Prolapse uterus, Internal and external haemorrhoids, Anal fistula, Vasectomy, Tubal pregnancy & Tubal ligation
AN48.6	Describe the neurological basis of Automatic bladder

AN48.7	Mention the lobes involved in benign prostatic hypertrophy & prostatic cancer
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AN48.8	Mention the structures palpable during vaginal & rectal examination
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Topic 49: Perineum

AN49.1	Describe & demonstrate the superficial & deep perineal pouch (boundaries and contents)
AN49.2	Describe & identify Perineal body
AN49.3	Describe & demonstrate Perineal membrane in male & female
AN49.4	Describe & demonstrate boundaries, content & applied anatomy of Ischioanal fossa
AN49.5	Explain the anatomical basis of Perineal tear, Episiotomy, Perianal abscess and Anal fissure

Topic 50: Vertebral column

AN50.1	Describe the curvatures of the vertebral column
AN50.2	Describe & demonstrate the type, articular ends, ligaments and movements of Intervertebral joints, Sacroiliac joints & Pubic symphysis
AN50.3	Describe lumbar puncture (site, direction of the needle, structures pierced during the lumbar puncture)
AN50.4	Explain the anatomical basis of Scoliosis, Lordosis, Prolapsed disc, Spondylolisthesis & Spina bifida

Topic 51: Sectional Anatomy

AN51.1	Describe & identify the cross-section at the level of T8, T10 and L1 (transpyloric plane)
AN51.2	Describe & identify the midsagittal section of male and female pelvis

Topic 52: Histology & Embryology

AN52.1	Describe & identify the microanatomical features of Gastro-intestina system: Oesophagus, Fundus of stomach, Pylorus of stomach, Duodenum, Jejunum, Ileum, Large intestine, Appendix, Liver, Gall bladder, Pancreas & Suprarenal gland
AN52.2	Describe & identify the microanatomical features of: Urinary system: Kidney, Ureter & Urinary bladder Male Reproductive System: Testis, Epididymis, Vas deferens, Prostate & penis Female reproductive system: Ovary, Uterus, Uterine tube, Cervix, Placenta & Umbilical cord
AN52.3	Describe & identify the microanatomical features of Cardiooesophageal junction, Corpus luteum
AN52.4	Describe the development of anterior abdominal wall
AN52.5	Describe the development and congenital anomalies of Diaphragm
AN52.6	Describe the development and congenital anomalies of: Foregut, Midgut & Hindgut
AN52.7	Describe the development of Urinary system
AN52.8	Describe the development of male & female reproductive system

Topic 53: Osteology

AN53.1	Identify & hold the bone in the anatomical position, Describe the salient features, articulations & demonstrate the attachments of muscle groups
AN53.2	Demonstrate the anatomical position of bony pelvis & show boundaries of pelvic inlet, pelvic cavity, pelvic outlet
AN53.3	Define true pelvis and false pelvis and demonstrate sex determination in male & female bony pelvis
AN53.4	Explain and demonstrate clinical importance of bones of abdominopelvic region (sacralization of lumbar vertebra, Lumbarization of 1st sacral vertebra, types of bony pelvis & Coccyx)

Topic 54: Radiodiagnosis

AN54.1	Describe the principles of Plain and contrast radiography, Computed Tomography, Magnetic Resonance Imaging, Positron Emission Tomography scan and Digital subtraction angiography
AN54.2	Describe & identify features of plain X ray abdomen
AN54.3	Describe & identify the special radiographs of abdominopelvic region (contrast X ray Barium swallow, Barium meal, Barium enema Cholecystography, Intravenous pyelography & Hysterosalpingography)
AN54.4	Describe role of ERCP, CT abdomen, MRI, Arteriography in radiodiagnosis of abdomen

Topic 55: Surface marking

- AN55.1** Demonstrate the surface marking of Regions and planes of abdomen, Superficial inguinal ring, Deep inguinal ring, McBurney's point, Renal Angle & Murphy's point
- AN55.2** Demonstrate the surface projections of: Stomach, Liver, Fundus of gal bladder, Spleen, Duodenum, Pancreas, Ileocaecal junction, Kidneys & Root of mesentery

Topic 56: Meninges & CSF

- AN56.1** Describe & identify various layers of meninges with its extent & modifications
- AN56.2** Describe formation, circulation and absorption of CSF with its applied anatomy.

Topic 57: Spinal Cord

- AN57.1** Identify external features of spinal cord
- AN57.2** Describe extent of spinal cord in child & adult with its clinical implication
- AN57.3** Draw & label transverse section of spinal cord at mid-cervical & mid-thoracic level
- AN57.4** Enumerate ascending & descending tracts at mid thoracic level of spinal cord

- AN57.5** Describe the anatomical basis of clinical conditions affecting the grey and white matter of spinal cord (Brown-Sequard Syndrome, Poliomyelitis Amyotrophic lateral sclerosis or motor neuron disease, Syringomyelia Hereditary sensory neuropathy, Subacute Combined degeneration Transversemyelitis, paraplegia)

Topic 58: Medulla Oblongata

- AN58.1** Identify external features of medulla oblongata
- AN58.2** Describe transverse section of medulla oblongata at the level of 1) pyramidal decussation, 2) sensory decussation 3) Inferior Olivary Nucleus
- AN58.3** Describe cranial nerve nuclei in medulla oblongata with their functional group
- AN58.4** Describe the anatomical basis of clinical conditions affecting the medulla oblongata (Medial and lateral medullary syndromes, Crossed Diplegia)

Topic 59: Pons

- AN59.1** Identify external features of pons
- AN59.2** Draw & label transverse section of pons at the upper and lower level
- AN59.3** Describe cranial nerve nuclei in pons with their functional group
- AN59.4** Describe the anatomical basis of clinical conditions affecting the pons (Locked-in syndrome, Pontine haemorrhage, Foville syndrome, Raymond syndrome, Millard-Gubler syndrome)

Topic 60: Cerebellum

AN60.1	Describe & demonstrate external & internal features of cerebellum
AN60.2	Describe connections of cerebellar cortex and intracerebellar nuclei
AN60.3	Describe anatomical basis of cerebellar dysfunction

Topic 61: Midbrain

AN61.1	Identify external & internal features of midbrain
AN61.2	Describe internal features of midbrain at the level of superior & inferior colliculus
AN61.3	Describe the anatomical basis of clinical conditions affecting the midbrain (Weber syndrome, Benedikt syndrome, Parinaud syndrome)

Topic 62: Cranial nerve nuclei & Cerebral hemispheres

AN62.1	Describe the cranial nerve nuclei with its functional components
AN62.2	Describe & demonstrate surfaces, sulci, gyri, poles, & functional areas of cerebral hemisphere. Also describe the effects of damage to various functional areas of cerebral cortex
AN62.3	Describe the white matter of cerebrum. Also describe the effects of damage to corpus callosum and different parts of internal capsule

AN62.4 Describe the parts & major connections of basal ganglia & limbic lobe. Also explain the anatomical basis of Parkinson's disease, chorea, athetosis and ballismus

AN62.5 Describe boundaries, parts, gross relations, major nuclei and connections of dorsal thalamus, hypothalamus, epithalamus, metathalamus and subthalamus

AN62.6 Describe & identify formation, branches & major areas of distribution of circle of Willis

Topic 63: Ventricular System & Special sensory pathways

AN63.1 Describe & demonstrate parts, boundaries & features of 3rd, 4th & lateral ventricle

AN63.2 Describe anatomical basis of congenital hydrocephalus

AN63.3 Describe the olfactory, visual, auditory and gustatory pathways

Topic 64: Histology & Embryology

AN64.1 Describe & identify the microanatomical features of Spinal cord, Cerebellum & Cerebrum

AN64.2 Describe the development of neural tube, spinal cord, medulla oblongata, pons, midbrain, cerebral hemisphere & cerebellum

AN64.3 Describe various types of open neural tube defects with its embryological basis

Topic 65: Epithelium histology

AN65.1 Identify epithelium under the microscope & describe the various types that correlate to its function

AN65.2 Describe the ultrastructure of epithelium

Topic 66: Connective tissue histology

AN66.1 Describe & identify various types of connective tissue with functional correlation

AN66.2 Describe the ultrastructure of connective tissue

Topic 67: Muscle histology

AN67.1 Describe & identify various types of muscle under the microscope

AN67.2 Classify muscle and describe the structure-function correlation of the same

AN67.3 Describe the ultrastructure of muscular tissue

Topic 68: Nervous tissue histology

AN68.1 Describe & Identify multipolar & unipolar neuron, ganglia, peripheral nerve under the microscope

AN68.2 Describe the structure-function correlation of neuron

AN68.3 Describe the ultrastructure of nervous tissue

Topic 69: Blood Vessels

AN69.1 Identify elastic & muscular blood vessels, capillaries under the microscope

AN69.2 Describe the various types and structure-function correlation of blood vessel

AN69.3 Describe the ultrastructure of blood vessels

Topic 70: Glands & Lymphoid tissue

AN70.1 Identify exocrine gland under the microscope & distinguish between serous, mucous and mixed acini

AN70.2 Identify the lymphoid tissue under the microscope & describe microanatomy of lymph node, spleen, thymus, tonsil and correlate the structure with function

Topic 71: Bone & Cartilage

AN71.1 Identify bone under the microscope; classify various types and describe the structure-function correlation of the same

AN71.2 Identify cartilage under the microscope & describe various types and structure- function correlation of the same

Topic 72: Integumentary System

AN72.1 Identify the skin and its appendages under the microscope and correlate the structure with function

Topic 73: Chromosomes

AN73.1 Describe the structure of chromosomes with classification

AN73.2 Describe technique of karyotyping with its applications

AN73.3 Describe the Lyon's hypothesis

Topic 74: Patterns of Inheritance

AN74.1	Describe mendelian and non-mendelian inheritance. Explain various modes of inheritance with examples.
AN74.2	Draw pedigree charts for the various types of inheritance & give examples of diseases of each mode of inheritance
AN74.3	Describe multifactorial inheritance with examples
AN74.4	Describe the genetic basis & clinical features of Achondroplasia, Cystic Fibrosis, Vitamin D resistant

Topic 75: Principles of Genetics, Chromosomal Aberrations & Clinical Genetics

AN75.1	Describe the structural and numerical chromosomal aberrations
AN75.2	Explain the terms mosaics and chimeras with example
AN75.3	Describe the genetic basis & clinical features of: Prader Willi syndrome, Edward syndrome, Patau syndrome, Down syndrome, Turner Syndrome & Klinefelter syndrome
AN75.4	Describe genetic basis of variation: polymorphism and mutation
AN75.5	Describe in brief: genetic counseling, karyotyping, FISH, PCR and genetic sequencing

Topic 76: Introduction to embryology

AN76.1	Describe the stages of human life
AN76.2	Explain the terms- phylogeny, ontogeny, trimester, viability

Topic 77: Gametogenesis and fertilization

AN77.1	Describe the uterine changes occurring during the menstrual cycle
AN77.2	Describe the synchrony between the ovarian and menstrual cycles
AN77.3	Describe spermatogenesis and oogenesis along with diagrams
AN77.4	Describe the stages and consequences of fertilisation
AN77.5	Describe the anatomical principles underlying contraception
AN77.6	Describe teratogenic influences: fertility and sterility, surrogate motherhood, social significance of “sex- ratio”.

Topic 78: Second week of development

AN78.1	Describe cleavage and formation of blastocyst
AN78.2	Describe the development of trophoblast
AN78.3	Describe the process of implantation & common abnormal sites of implantation
AN78.4	Describe the formation of extra-embryonic mesoderm and coelom, bilaminar disc and prochordal plate
AN78.5	Describe abortion, decidual reaction, pregnancy test

Topic 79: 3rd to 8th week of development

AN79.1	Describe the formation & fate of the primitive streak
AN79.2	Describe formation & fate of notochord
AN79.3	Describe the process of neurulation

AN79.4	Describe the development of somites and intra-embryonic coelom
AN79.5	Explain embryological basis of congenital malformations, nucleus pulposus, sacrococcygeal teratomas, neural tube defects
AN79.6	Describe the diagnosis of pregnancy in first trimester and role of teratogens, alpha-fetoprotein

Topic 80: Fetal membranes

AN80.1	Describe formation, functions & fate of chorion, amnion, yolk sac, allantois & decidua
AN80.2	Describe formation & structure of umbilical cord
AN80.3	Describe formation of placenta, its physiological functions, foetomaternal circulation & placental barrier
AN80.4	Describe embryological basis of twinning in monozygotic & dizygotic twins
AN80.5	Describe role of placental hormones in uterine growth & parturition
AN80.6	Explain embryological basis of estimation of fetal age.
AN80.7	Describe various types of umbilical cord attachments

Topic 81: Prenatal Diagnosis

AN81.1	Describe various invasive & non-invasive methods of prenatal diagnosis
AN81.2	Describe indications, process and disadvantages of amniocentesis

AN81.3	Describe indications, process and disadvantages of chorion villus biopsy
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Topic 82: Ethics in Anatomy

AN82.1	Demonstrate respect, and follow the correct procedure when handling cadavers and other biologic tissue
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PHYSIOLOGY

12 topics · 136 competencies · Phase I MBBS

Subject Goals

At the end of physiology teaching, the learner must be able to:

- i. Demonstrate knowledge of normal human physiology, organizational and functional relationship between cells, tissues and organs and body systems, age and sex related physiological changes in the organ functions that reflect normal growth and development.
- ii. Explain physiological variations (Genotype/Phenotype) with healthy ageing through the course of life i.e. fetal, neonatal, childhood, adolescence and adulthood and demonstrate understanding of the physiological responses and adaptation to environment and exercise.
- iii. Perform experiments to demonstrate physiological phenomenon and principles, interpret investigation results falling within the scope of physiology.
- iv. Apply principles of Physiology in clinicopathological conditions, diagnosis, investigations and management of diseased conditions.
- v. Conduct physical examination (general and system based) of a normal subject in real or simulated conditions and demonstrate understanding of altered findings in physical examination of diseased conditions.

Topics and Competencies

Topic 1: General Physiology

PY1.1	Describe the structure and functions of a cell, intercellular communication and their applications in Clinical care and research
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PY1.2	Discuss the principles of homeostasis and feedback mechanism
PY1.3	Describe apoptosis (programmed cell death) , explain its mechanism of action and physiological significance.
PY1.4	Describe and discuss various transport mechanisms across cell membranes
PY1.5	Describe the fluid compartments of the body, its ionic composition & measurement methods
PY1.6	Describe the concept of pH & Buffer systems in the body
PY1.7	Describe the molecular basis of resting membrane potential (RMP) and generation of action potential in a nerve fiber

Topic 2: Haematology

PY2.1	Describe the composition and functions of blood and its components
PY2.2	Discuss the origin, forms, variations and functions of plasma proteins and its clinical implications
PY2.3	Describe the physiological structure, synthesis , functions and breakdown of Hemoglobin. Discuss its variants and clinical significance.
PY2.4	Describe Erythropoiesis & discuss its regulation in physiological and pathological situations
PY2.5	Describe anaemias, polycythemia & jaundice and discuss its physiological principles of management

PY2.6	Describe the formation of WBC (Leucopoiesis), structure and function of various WBC types and their regulatory mechanisms
PY2.7	Discuss 'Immunity' in terms of its types, development, regulation and physiological significance
PY2.8	Describe the formation of platelets (thrombopoiesis), structure, functions and variations.
PY2.9	Describe hemostasis, coagulation pathways, mechanism of action of anticoagulants and briefly discuss pathophysiological aspects of bleeding & clotting disorders (e.g. hemophilia, purpura)
PY2.10	Discuss types of blood groups, clinical importance of blood grouping, blood banking and transfusion
PY2.11	Estimate Hb, RBC, TLC, DLC, Blood groups, BT/CT, RBC indices
PY2.12	Describe the test to measure Erythrocyte Sedimentation Rate (ESR), Osmotic fragility, Hematocrit, and interpret its findings
PY2.13	Describe steps for reticulocyte and platelet count

Topic 3: Nerve and Muscle Physiology

PY3.1	Describe the structure and functions of a neuron and neuroglia; Discuss nerve growth factors
PY3.2	Describe the types, functions, properties of nerve fibers including strength duration curve, chronaxie and rheobase
PY3.3	Classify nerve injury and discuss the mechanism of degeneration and regeneration in peripheral nerves

PY3.4	Describe the microscopic structure of neuro-muscular junction (NMJ) and mechanism of neuromuscular transmission
PY3.5	Discuss the applied aspects of neuromuscular junction : myasthenia gravis, Lambert Eaton syndrome and neuromuscular blocking agents.
PY3.6	Describe the different types of muscle fibres, their structure and physiological basis of action potential
PY3.7	Describe properties, action potential and molecular basis of muscle contraction in skeletal muscle
PY3.8	Describe properties, action potential and molecular basis of muscle contraction in smooth muscle
PY3.9	Describe the mode of muscle contraction (isometric and isotonic), energy source, muscle metabolism and gradation of muscular activity
PY3.10	Enumerate and briefly discuss myopathies
PY3.11	Perform Ergography and calculate the work done by a skeletal muscle
PY3.12	Observe with Computer assisted learning (i) Amphibian nerve -muscle experiments (ii) Amphibian cardiac experiments

Topic 4: Gastro-intestinal Physiology

PY4.1	Describe the functional anatomy of digestive system
PY4.2	Enumerate various Gastrointestinal hormones (GI) hormones, discuss their functions and regulation

PY4.3	Describe the composition, mechanism of secretion, functions, and regulation of saliva
PY4.4	Describe the composition, mechanism of secretion, functions, and regulation of gastric juice. Discuss various gastric function tests
PY4.5	Describe the composition, mechanism of secretion, functions, and regulation of pancreatic juice including various pancreatic exocrine function tests
PY4.6	Describe the composition, mechanism of secretion, functions, and regulation of intestinal juices
PY4.7	Describe the physiology of digestion and absorption of nutrients
PY4.8	Describe GIT movements, its regulation and physiological significance including defecation reflex and the role of dietary fibers
PY4.9	Describe the structure, functions and secretion of liver and gallbladder with elaboration of various liver function tests
PY4.10	Describe the Gut-Brain Axis and its physiological significance
PY4.11	Discuss (in brief) the applied physiology of GIT viz. Peptic ulcer, gastroesophageal reflux disease, vomiting, diarrhoea, constipation, Adynamic ileus, Hirschsprung's disease
PY4.12	Obtain relevant history and conduct correct General and Clinical examination of the abdomen in a normal volunteer or simulated environment

Topic 5: Cardiovascular Physiology

PY5.1	Describe the functional anatomy of heart including chambers and coronary circulation
PY5.2	Describe the properties of cardiac muscle including its morphology, electrical, mechanical and metabolic functions
PY5.3	Describe generation and conduction of cardiac impulse along with the conduction pathway (including pacemaker potential).
PY5.4	Discuss the physiological events occurring during the cardiac cycle, concurrent pressure volume changes, generation of heart sounds and murmur
PY5.5	Describe the physiology of electrocardiogram (E.C.G), the cardiac axis and its applications
PY5.6	Discuss physiological variations in ECG waveforms, abnormal waveforms and intervals, arrhythmias, heart blocks and myocardial Infarction
PY5.7	Discuss haemodynamics of circulatory system
PY5.8	Describe and discuss local and systemic cardiovascular regulatory mechanisms
PY5.9	Describe heart rate, factors affecting heart rate, and its regulation
PY5.10	Describe cardiac output, factors affecting cardiac output and its regulation.
PY5.11	Describe blood pressure, factors affecting blood pressure and its regulation

PY5.12	Describe & discuss regional circulation including microcirculation, lymphatic circulation, cerebral, capillary, skin, foetal, pulmonary and splanchnic circulation
PY5.13	Describe the patho-physiology of shock, syncope heart failure with physiological basis of its management
PY5.14	Record blood pressure & pulse at rest and in different grades of exercise and postures in a volunteer or simulated environment
PY5.15	Record and interpret normal ECG in a volunteer or simulated environment
PY5.16	Obtain relevant history and conduct General and Clinical examination of the cardiovascular system in a normal volunteer or simulated environment

Topic 6: Respiratory Physiology

PY6.1	Describe the functional anatomy of respiratory tract and non-respiratory functions of lungs
PY6.2	Describe the mechanics of normal respiration, pressure changes during ventilation, lung volume and capacities (Static and Dynamic)
PY6.3	Describe the alveolar surface tension, compliance, airway resistance, ventilation, V/P ratio, diffusion capacity of lungs
PY6.4	Discuss the transport of respiratory gases viz Oxygen and Carbon dioxide across lungs and whole body
PY6.5	Describe the chemoreceptors (peripheral and central) and neural centres of respiration including chemical and neural regulation of respiration

PY6.6	Describe and discuss the pathophysiology of dyspnoea, hypoxia, cyanosis, asphyxia, drowning, periodic breathing and oxygen therapy
PY6.7	Discuss various lung function tests and their clinical significance in obstructive and restrictive lung diseases
PY6.8	Discuss the physiology of high altitude and acclimatization
PY6.9	Discuss the physiology of deep sea diving and decompression sickness
PY6.10	Perform Spirometry and interpret the findings (Digital / Manual)
PY6.11	Describe principles and methods of artificial respiration
PY6.12	Obtain relevant history and conduct correct General and Clinical examination of the respiratory system in a normal volunteer or simulated environment
PY6.13	Demonstrate the correct technique to perform measurement of peak expiratory flow rate in a normal volunteer or simulated environment

Topic 7: Renal Physiology

PY7.1	Describe the functional anatomy of kidney and non-excretory functions of kidney
PY7.2	Describe the structure and functions of juxta glomerular apparatus and role of renin-angiotensin system
PY7.3	Describe the mechanism of urine formation involving processes of filtration (Glomerular filtration), tubular reabsorption & secretion.

PY7.4	Describe the mechanism of urine concentration and dilution (Counter current Multiplier & Exchanger)
PY7.5	Describe the renal regulation of fluid and electrolytes & acid-base balance
PY7.6	Describe the innervations of urinary bladder, physiology of micturition and its abnormalities
PY7.7	Describe cystometry and discuss the normal cystometrogram
PY7.8	Discuss various Renal Function Tests with its physiological significance and clinical implication of Renal clearance
PY7.9	Discuss the role of artificial kidneys, dialysis and indications of renal transplant

Topic 8: Endocrine Physiology

PY8.1	Describe the functional anatomy of endocrine glands, mechanism of hormonal action (steroid and peptide) and hypothalamus pituitary axis {HPA}
PY8.2	Describe the synthesis, secretion, transport, physiological actions, regulation and effect of altered (hypo and hyper) secretion of pituitary gland
PY8.3	Describe the synthesis, secretion, transport, physiological actions, regulation and effect of altered (hypo and hyper) secretion of thyroid gland including thyroid function tests
PY8.4	Describe the synthesis, secretion, transport, physiological actions, regulation and effect of altered (hypo and hyper) secretion of adrenal gland and its function tests

PY8.5	Describe the synthesis, secretion, transport, physiological actions, regulation and effect of altered (hypo and hyper) secretion of parathyroid gland with emphasis of physiology of bone and calcium metabolism
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PY8.6	Describe the synthesis, secretion, transport, physiological actions, regulation and effect of altered (hypo and hyper) secretion of pancreatic gland including pancreatic function tests
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PY8.7	Describe the physiology of Thymus & Pineal Gland
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Topic 9: Reproductive Physiology

PY9.1	Explain sex determination, sex differentiation and their abnormalities and discuss the effects of removal of gonads on physiological functions
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PY9.2	Describe and discuss puberty: onset, progression, stages; early and delayed puberty.
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PY9.3	Describe the functional anatomy of male reproductive system, functions of testis, spermatogenesis and discuss the functions and regulations of testosterone hormone
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PY9.4	Describe the functional anatomy of female reproductive system: functions of ovary and its hormones (estrogen and progesterone) ; hormonal regulation by hypothalamic pituitary gonadal (HPG axis)
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PY9.5	Discuss the menstrual cycle, uterine and ovarian changes, hormonal regulation and its implications in reproductive physiology
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PY9.6	Enumerate male and female contraceptive methods, rationale of its prescription, side effects and its advantages & disadvantages
PY9.7	Discuss the physiology of pregnancy, parturition & lactation.
PY9.8	Discuss the physiological basis of various pregnancy tests
PY9.9	Discuss the hormonal changes and their effects during perimenopause and menopause
PY9.10	Discuss the common causes of infertility in a couple and role of IVF in managing a case of infertility

Topic 10: Central Nervous System Physiology

PY10.1	Describe and discuss the functional organization of central nervous system (brain and spinal cord)
PY10.2	Describe the functional anatomy of peripheral nervous system (including autonomic nervous system)
PY10.3	Classify the neurotransmitters and discuss the chemical transmission in the nervous system.
PY10.4	Discuss the classification, functions and properties of synapse
PY10.5	Discuss the classification, functions and properties of reflex
PY10.6	Discuss the classification, functions and properties of receptors
PY10.7	Discuss somatic sensations, ascending tracts, (sensory tracts) and applied aspects of sensory system

PY10.8	Discuss Physiology of pain including pain pathways and its modulation with special emphasis on gate control theory of pain
PY10.9	Describe the course of descending tracts (pyramidal and extra pyramidal), its clinical implications including difference in Upper motor neuron (UMN) and lower motor neuron (LMN) lesions
PY10.10	Discuss types and clinical features of spinal cord lesions (complete, incomplete transection and hemisection - Brown Sequard syndrome)
PY10.11	Describe functional anatomy of cerebellum, its connections, functions and clinical abnormalities .
PY10.12	Discuss functional anatomy of basal ganglia , its connections, functions and Clinical abnormalities .
PY10.13	Discuss the mechanism of maintenance of tone, posture and control of body movements
PY10.14	Discuss functional anatomy of thalamus , its connections, functions and clinical abnormalities .
PY10.15	Discuss functional anatomy of hypothalamus and limbic system , its connections, functions and clinical abnormalities .
PY10.16	Discuss functional anatomy of cerebral cortex, its connections, functions and Clinical abnormalities
PY10.17	Discuss the structure and functions of reticular activating system, sleep physiology and EEG waveforms during sleep wake cycle

PY10.18	Discuss the physiological basis of memory, learning and speech and clinical alterations in speech
PY10.19	Obtain relevant history and conduct correct General and Clinical examination of the nervous system: Higher functions, sensory system, motor system, reflexes in a normal volunteer or simulated environment
PY10.20	Obtain relevant history and conduct correct General and Clinical examination of the cranial nerves in a normal volunteer or simulated environment

Topic 11: Special Senses

PY11.1	Describe and discuss physiology of smell and its applied aspects
PY11.2	Describe and discuss physiology of taste sensation and applied aspects
PY11.3	Describe and discuss functional anatomy of ear and auditory pathways, vestibular apparatus and equilibrium
PY11.4	Discuss physiology of hearing, pathophysiology of deafness and hearing tests
PY11.5	Discuss functional anatomy of eye, visual pathway, light and pupillary reflex and clinical implication of lesions in visual pathway
PY11.6	Discuss physiology of image formation, refractive errors and physiological principles of its management
PY11.7	Discuss physiology of vision including colour vision and colour blindness

Topic 12: Integrated Physiology

PY12.1	Describe physiological mechanism of temperature regulation
PY12.2	Discuss adaptation to altered temperature (heat and cold) and mechanism of fever, cold injuries and heat stroke
PY12.3	Discuss cardio-respiratory and metabolic adjustments during exercise (isometric and isotonic), effects of physical training under different environmental conditions (heat and cold)
PY12.4	Discuss physiological consequences of sedentary lifestyle; metabolic and endocrinal consequences of obesity & metabolic syndrome.
PY12.5	Describe physiology of Infancy, Interpret growth charts and anthropometric assessment of infants
PY12.6	Describe and discuss physiology of aging, role of free radicals and antioxidants
PY12.7	Discuss the concept, criteria for diagnosis of Brain death and its implications
PY12.8	Discuss the physiology of yoga and meditation
PY12.9	Obtain history and perform general examination in the volunteer / simulated environment
PY12.10	Demonstrate Basic Life Support in a simulated environment

BIOCHEMISTRY

14 topics · 84 competencies · Phase I MBBS

Subject Goals

The learner after teaching–learning in Biochemistry should be able to:

- i. Understand and explain Biochemical and molecular processes involved in health and disease.
- ii. Enlist and describe the cell organelles with their molecular and functional organization.
- iii. Understand basic enzymology and emphasise its clinical applications wherein regulation of enzymatic activity is disturbed.
- iv. Describe the importance of nutrition in health and disease.
- v. Describe digestion and assimilation of nutrients and consequences of malnutrition.
- vi. Describe function and interrelationships of various biomolecules and consequences of deviation from the normal.
- vii. Describe and integrate metabolic pathways of various biomolecules with their regulatory mechanisms relevant to clinical conditions.
- viii. Describe the biochemical basis and rationale of clinical laboratory tests, perform biochemical analytical tests relevant to clinical screening and diagnosis using conventional techniques/instruments and interpret investigative data.
- ix. Explain the biochemical basis of inherited disorders with their associated sequel.
- x. Describe mechanisms involved in maintenance of water, electrolyte and acid–base balance and consequences of their imbalances.

- xi. Outline basics of genetics, explain the molecular mechanisms of gene expression and regulation, basic principles of biotechnology and latest techniques and their applications in medicine.
- xii. Demonstrate the skills of solving scientific and clinical problems and decision making.

Topics and Competencies

Topic 1: Basic Biochemistry

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| BC1.1 | Describe the molecular and functional organization of a cell and its sub-cellular components and composition and functions of biological membranes. |
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Topic 2: Enzyme

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| BC2.1 | Explain fundamental concepts of enzyme, isoenzyme and coenzyme. Enumerate the main classes of IUBMB nomenclature. |
| BC2.2 | Describe and explain the basic principles of enzyme activity |
| BC2.3 | Describe and discuss enzyme Inhibition and role of enzymes or drugs as Inhibitors, and enzymes as therapeutic agents. |
| BC2.4 | Describe and discuss the clinical utility of various serum enzymes in laboratory and their use as markers of various pathological conditions. |
| BC2.5 | Interpret laboratory results of enzymes in various disorders. |

Topic 3: Chemistry and Metabolism of Carbohydrates

BC3.1	Discuss and differentiate monosaccharides, di-saccharides and polysaccharides with examples, their importance as energy fuel structural element, and storage molecule in human body.
BC3.2	Describe the digestion, absorption and transport of carbohydrates from food along with its disorders.
BC3.3	Define and briefly describe the pathways of carbohydrate metabolism and their regulation (glycolysis, gluconeogenesis, TCA, and significance of glycogen metabolism and HMP shunt), with associated disorders.
BC3.4	Describe and discuss the regulation, functions and integration of minor Carbohydrate Metabolism pathway briefly along with associated diseases /disorders.
BC3.5	Discuss the mechanism and significance of blood glucose regulation (Glucose homeostasis) in health and disease. Describe the types, Biochemical changes, complications and laboratory investigations related to diabetes & other carbohydrate metabolic disorders.
BC3.6	Interpret the results of analytes associated with metabolism of carbohydrates and other laboratory investigations related to disorders of carbohydrate metabolism.

Topic 4: Chemistry and Metabolism of Lipids

BC4.1	Describe and discuss main classes of lipids and their functions.
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BC4.2	Describe the digestion and absorption of dietary lipids and its associated disorders.
BC4.3	Describe and discuss the fatty acid oxidation, metabolism of ketone bodies along with their clinical significance.
BC4.4	Describe metabolism of Triglycerides and cholesterol metabolism along with its regulation and clinical significance.
BC4.5	Describe the metabolism of lipoproteins with brief overview of lipoprotein structure, their interrelations & relations with atherosclerosis.
BC4.6	Discuss Biological role and therapeutic applications of Eicosanoids and their Inhibitors.
BC4.7	Describe Fatty liver, cholelithiasis and obesity.
BC4.8	Interpret laboratory results of analytes associated with metabolism of lipids

Topic 5: Chemistry & Metabolism of Proteins and Immunology

BC5.1	Discuss briefly structure of amino acids and classify amino acids on the basis of Nutritional and Metabolic significance.
BC5.2	Discuss classification of proteins, structural organization, functions and clinical aspects.
BC5.3	Describe the digestion and absorption of dietary proteins
BC5.4	Describe plasma proteins and their functions and brief overview of normal and abnormal electrophoretic pattern of serum proteins, acute phase proteins.

BC5.5	Describe the structure, functions and disorders of Immunoglobulins with brief description of cellular and humoral Immunity.
BC5.6	Describe the formation, transport, detoxification of Ammonia, Ammonia toxicity and its clinical significance.
BC5.7	Describe the specialized products formed from the amino acids Glycine, Phenylalanine, Tyrosine, Tryptophan, and Methionine, branched chain amino acids and Arginine and the inborn errors associated with them Discuss new-born screening.
BC5.8	Describe the structure and functions of haem in the body and describe the processes involved in its metabolism with emphasis on jaundice and describe porphyrin metabolism.
BC5.9	Describe the major types of Hemoglobin and its types, derivatives & variants found in the body and their physiological / pathological relevance

Topic 6: Extracellular Matrix

BC6.1	Enumerate the functions and components of the extracellular matrix (ECM).
BC6.2	Discuss the involvement of ECM components in health and disease.
BC6.3	Describe protein targeting & sorting along with its associated disorders.

Topic 7: Integration of Metabolism and Biological Oxidation

BC7.1	Describe the integration of various metabolic processes in the body (Carbohydrate, Lipid, and Protein).
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BC7.2	Describe the Biochemical processes involved in generation of energy in cells.
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Topic 8: Vitamins and Nutrition

BC8.1	Describe the Biochemical role of vitamins in the body and explain the manifestations of their deficiency
BC8.2	Discuss the importance of various dietary components and explain importance of dietary fibre.
BC8.3	Describe the types and causes of protein energy malnutrition and its effects.
BC8.4	Provide dietary advice for optimal health in childhood and adult in disease conditions like diabetes mellitus, coronary artery disease and in pregnancy.
BC8.5	Describe the causes (including dietary habits), effects and health risks associated with being overweight/ obese / metabolic syndrome
BC8.6	Summarize the nutritional importance of commonly used items of food including fruits and vegetables (macro-molecules & its importance).

Topic 9: Minerals, electrolytes, Water and Acid base balance

BC9.1	Describe the dietary sources, absorption, transport, and metabolism, Biochemical functions of Iron, Calcium and copper with its associated clinical disorders.
BC9.2	Discuss Magnesium, Zinc and Phosphorus along with its clinical significance and discuss the functions of trace elements

BC9.3 Describe the processes involved in maintenance of normal pH, water & electrolyte balance of body fluids and the derangements associated with them

Topic 10: Molecular Biology

BC10.1 Describe nucleotides and nucleic acids and their clinical significance.

BC10.2 Describe briefly synthesis of purines in the body with special stress on salvage pathway.

BC10.3 Describe the degradation of purines and its significance with associated disorders.

BC10.4 Describe in brief the major steps involved in Replication, Transcription, and translation.

BC10.5 Describe the types of DNA repair, gene mutations and associated disorders.

BC10.6 Describe basic mechanism of regulation of gene expression

BC10.7 Describe applications of molecular technologies like recombinant DNA technology and PCR in the diagnosis and treatment of diseases. Briefly discuss microarray, FISH, CRISPR

Topic 11: Organ Function tests and Hormones

BC11.1 Describe the function tests of kidney, liver, thyroid and adrenal glands and their clinical significance. Interpret the function tests report.

BC11.2 Enumerate the hormones and markers related to reproduction and reproductive health and their clinical interpretation (For e.g. LH, FSH, Prolactin, beta-HCG, Estrogen Progesterone, testosterone and AMH. Discuss importance of prenatal screening.

Topic 12: Xenobiotic, oxidative stress and antioxidants

BC12.1 Describe the role of xenobiotics in disease in health and disease

BC12.2 Describe the anti-oxidant defense systems in the body.

BC12.3 Describe the role of oxidative stress in the pathogenesis of conditions such as cancer, complications of diabetes mellitus and atherosclerosis

Topic 13: Miscellaneous

BC13.1 Describe oncogenesis, oncogenes & its activation with focus on p53 & apoptosis.

BC13.2 Describe various Biochemical tumor markers and the Biochemical basis of cancer therapy.

BC13.3 Discuss briefly on HIV and Biochemical changes in AIDS.

BC13.4 Discuss metabolism of alcohol with Biochemical changes and effects of chronic alcoholism.

BC13.5 Describe the role of Artificial Intelligence in clinical Biochemistry laboratory practices.

Topic 14: Biochemical Laboratory test / Practical

BC14.1	Describe commonly used laboratory apparatus equipments, good / safe laboratory practice, Biomedical hazards & waste management.
BC14.2	Describe estimation of pH by pH meter or ABG analyser and interpretation of results with paper case scenarios.
BC14.3	Describe the physical properties, chemical constituents of normal urine and abnormal constituents of urine and Perform urine analysis to determine normal and abnormal constituents (including dipsticks method demonstration).
BC14.4	Identify abnormal constituents in urine, interpret the findings and correlate these with pathological states and prepare a urine report.
BC14.5	Describe screening of urine for inborn errors & describe the use of paper chromatography
BC14.6	Describe the principles of Colorimetry & Spectrophotometry.
BC14.7	Perform estimation of glucose by manual / semi-automated analyzer method and demonstrate glucometer usage. and interpretation of results with clinical scenarios.
BC14.8	Perform estimation of urea and calculate BUN and interpretation of results in clinical scenarios.
BC14.9	Perform the estimation of serum creatinine and calculate creatinine clearance.
BC14.10	Perform estimation of uric acid in serum and interpretation of results with clinical scenarios.

BC14.11	Perform estimation of serum proteins, albumin and A:G ratio
BC14.12	Perform the estimation of serum total cholesterol
BC14.13	Perform the estimation of serum Bilirubin by manual / semi-automated analyzer method.
BC14.14	Describe estimation of calcium and phosphorus and interpretation of results.
BC14.15	Describe the estimation Triglycerides, HDL and calculation of LDL and interpretation of results with clinical scenarios.
BC14.16	Describe the estimation of SGOT (AST) / SGPT (ALT) / Alkaline Phosphatase and interpretation of results with clinical scenarios.
BC14.17	Describe briefly various body fluids & discuss the composition of CSF.
BC14.18	Observe use of commonly used equipments/techniques in Biochemistry laboratory including: pH meter Paper chromatography of amino acid Protein electrophoresis
BC14.19	Explain the basis and rationale of Biochemical tests done and interpretation of laboratory results in the following conditions: Diabetes mellitus, Obesity, dyslipidaemia, Fatty liver myocardial infarction, Renal failure, Gout, Nephrotic syndrome, Jaundice, Liver diseases, pancreatitis, disorders of acid- base balance, Thyroid disorders, Genetic disorders Nutritional disorders Vitamin deficiency disorders, Disorders of Mineral metabolism, Disorders of electrolyte metabolism.
BC14.20	Describe & Identify Pre-Analytical (especially order of draw, tourniquet technique), Analytical, Post Analytical errors.

BC14.21	Describe Quality control and identify basic L J charts in Clinical biochemistry lab.
BC14.22	Describe performance of OGTT, Glucose Challenge Test and HbA1c and interpretation of results with clinical scenarios.
BC14.23	Calculate energy content of different food Items, identify food items with high and low glycaemic index and explain the importance of these in the diet.
BC14.24	Observe, Interpret and discuss the baseline, diagnostic, prognostic, and discharge investigations of clinical biochemistry.

COMMUNITY MEDICINE

20 topics · 136 competencies · Phase I MBBS

Subject Goals

At the end of teaching–learning in Community Medicine, the student should be able to:

- i. Demonstrate understanding of the role of the primary care physician for preventive, promotive, curative, rehabilitative, palliative care and referral services.
- ii. Demonstrate understanding of the concept of health and disease, demography, population dynamics and disease burden in national and global context, and comprehend principles of health economics and hospital management.
- iii. Apply the understanding of physical, social, psychological, economic and environmental determinants of health and disease, and recognise and manage common health problems including physical, emotional and social aspects at individual, family and community level in the context of National Health Programmes.
- iv. Implement and monitor National Health Programmes in the primary care setting; recognise, investigate, report, plan and manage community health problems including malnutrition and emergencies.
- v. Apply understanding of the role of nutrition in health promotion and disease prevention.
- vi. Demonstrate the role of researcher and community medicine physician by understanding epidemiological study designs and their application, epidemiology of diseases, and the ability to critically review.
- vii. Demonstrate understanding of pandemic and epidemic situations with emerging and re-emerging diseases and be able to investigate under supervision and plan, advise and promote preventive aspects as per

international and national health regulations and programmes.

- viii. Demonstrate understanding of all principles of public health, community medicine and preventive and social aspects, utilising the Family Adoption Programme, providing services to the families adopted and being first-care physician under the guidance of a mentor.
- ix. Apply the principles of behaviour change communication for improving health-related aspects for communicable and non-communicable diseases, health-promotive aspects related to addictions, and health-related information and misinformation.

Topics and Competencies

Topic 1: Concept of Health and Disease

CM1.1	Define and describe the concept of Public Health
CM1.2	Define health; describe the concept of holistic health including concept of spiritual health and the relativeness & determinants of health
CM1.3	Describe the characteristics of agent, host and environmental factors in health and disease and the multifactorial etiology of disease
CM1.4	Describe and discuss the natural history of disease
CM1.5	Describe the application of interventions at various levels of prevention
CM1.6	Describe and discuss the concepts, the principles of Health promotion and Education, IEC and Behavioral change communication (BCC)
CM1.7	Enumerate and describe health indicators

CM1.8	Describe the Demographic profile of India and discuss its Impact on health
CM1.9	Demonstrate the role of effective Communication skills in health in a simulated environment
CM1.10	Demonstrate the Important aspects of the doctor patient relationship in a simulated environment

Topic 2: Relationship of social and behavioural to health and disease

CM2.1	Describe the steps and perform clinico socio-cultural and demographic assessment of the individual, family and community
CM2.2	Describe the socio-cultural factors, family (types), its role in health and disease & demonstrate in a simulated environment the correct assessment of socio-economic status
CM2.3	Describe and demonstrate in a simulated environment the assessment of barriers to good health and health seeking behavior
CM2.4	Describe social psychology, community behaviour and community relationship and their impact on health and disease
CM2.5	Describe poverty and social security measures and its relationship to health and disease

Topic 3: Environmental Health Problems

CM3.1	Describe the health hazards of air, water, noise, radiation and pollution
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CM3.2	Describe concepts of safe and wholesome water, sanitary sources of water, water purification processes, water quality standards, concepts of water conservation and rainwater harvesting
CM3.3	Describe the aetiology and basis of water borne diseases /jaundice /hepatitis /diarrheal diseases
CM3.4	Describe the concept of solid waste, human excreta and sewage disposal
CM3.5	Describe the standards of housing and the effect of housing on health
CM3.6	Describe the role of vectors in the causation of diseases. Also discuss National Vector Borne disease Control Program
CM3.7	Identify and describe the identifying features and life cycles of vectors of Public Health Importance and their control measures
CM3.8	Describe the mode of action, application cycle of commonly used insecticides and rodenticides

Topic 4: Principles of health promotion and education

CM4.1	Describe various methods of health education with their advantages and limitations
CM4.2	Describe the methods of organizing health promotion and education and counselling activities at individual family and community settings
CM4.3	Demonstrate and describe the steps in evaluation of health promotion and education program

CM4.4 Conduct a health education session for community awareness in a simulated environment/FAP/clinical posting

Topic 5: Nutrition

CM5.1 Describe the common sources of various nutrients and special nutritional requirements according to age, sex, activity, physiological conditions

CM5.2 Describe and demonstrate the correct method of performing a nutritional assessment of individuals, families and the community by using the appropriate method

CM5.3 Define and describe common nutrition related health disorders (including macro-PEM, Micro-iron, Zn, iodine, Vit.A), their control and management

CM5.4 Plan and recommend a suitable diet for the individuals and families based on local availability of foods and economic status, etc in a simulated environment

CM5.5 Describe the methods of nutritional surveillance principles nutritional education and rehabilitation in the context of socio- cultural factors.

CM5.6 Enumerate and discuss the National Nutrition Policy, important national nutritional Programs including the Integrated Child Development Services Scheme (ICDS) etc

CM5.7 Describe food hygiene

CM5.8 Describe and discuss the importance and methods of food fortification and effects of additives and adulteration

CM5.9 Perform nutritional assessment of individual, family and community using appropriate method and plan a diet for health promotion based on the assessment

CM5.10 Recommend a dietary plan for a person with DM/ HTN/ Obesity in a simulated environment/FAP/Clinical posting

CM5.11 Plan a diet for an adult which meets the protein (macro nutrients) requirements as per latest RDA-NIN guidelines for vegetarian/ ovo-vegetarian/non-vegetarian

CM5.12 Demonstrate different types of breastfeeding holds, latching, manual expression of breast milk using a baby model and breast model.

CM5.13 Counsel a mother about complementary feeding for different age groups of the child covering the 8 dietary diversity food groups, quantity, frequency, consistency of the food.

CM5.14 Demonstrate an awareness of their own personal health and nutrition

CM5.15 Demonstrate knowledge of the role of nutrition in health promotion and disease prevention

CM5.16 Have knowledge of breast feeding and complementary feeding Practices

CM5.17 Ability to counsel mothers on breast feeding with focus on attachment to breast and correct position of the newborn

CM5.18 Ability to counsel mothers on complementary feeding using National guidelines while being sensitive of cultural and socioeconomic influences

CM5.19	Assess the nutritional content of processed foods learning to understand labels, and empower patients to make informed nutritional decisions.
CM5.20	Counsel for diet modification for a diabetic/hypertensive/obese individual
CM5.21	Plan and conduct a health education session on nutrition in NCD clinic / in community
CM5.22	Counsel mother on breast feeding and complementary feeding

Topic 6: Basic statistics and its applications

CM6.1	Formulate research question for a study
CM6.2	Describe and discuss the principles and demonstrate the methods of collection, classification, analysis, interpretation and presentation of statistical data
CM6.3	Describe, discuss and demonstrate the application of elementary statistical methods including test of significance in various study designs
CM6.4	Enumerate, discuss and demonstrate Common sampling techniques, simple statistical methods, frequency distribution, measures of central tendency and dispersion
CM6.5	Able to understand use of statistical software for the data analysis
CM6.6	Perform descriptive statistics of a given data-set and interpret

Topic 7: Epidemiology

CM7.1	Define Epidemiology and describe and enumerate the principles, concepts and uses
CM7.2	Enumerate, describe and discuss the modes of transmission and measures for prevention and control of communicable and non-communicable diseases
CM7.3	Enumerate, describe and discuss the sources of epidemiological data
CM7.4	Define, calculate and interpret morbidity and mortality indicators based on given set of data
CM7.5	Enumerate, define, describe and discuss epidemiological study designs
CM7.6	Enumerate and evaluate the need of screening tests
CM7.7	Describe and demonstrate the steps in the Investigation of an epidemic of communicable disease and describe the principles of control measures
CM7.8	Describe the principles of association, causation and biases in epidemiological studies
CM7.9	Describe and demonstrate the application of computers in epidemiology
CM7.10	Able to demonstrate development of research proposal
CM7.11	Able to demonstrate the skills for critically appraise the research articles or research data

Topic 8: Epidemiology of communicable and non- communicable diseases

CM8.1	Describe and discuss the epidemiological and control measures including the use of essential laboratory tests at the primary care level for communicable diseases
CM8.2	Describe and discuss the epidemiological and control measures including the use of essential laboratory tests at the primary care level for Non-Communicable diseases (diabetes, Hypertension, Stroke, obesity and cancer etc.)
CM8.3	Enumerate and describe disease specific National Health Programs including their prevention and treatment of a case
CM8.4	Describe the principles and enumerate the measures to control a disease epidemic
CM8.5	Describe and discuss the principles of planning, Implementing and evaluating control measures for disease at community level bearing in mind the public health importance of the disease
CM8.6	Educate and train health workers in disease surveillance, control & treatment and health education
CM8.7	Describe the principles of management of information systems

Topic 9: Demography and vital statistics

CM9.1	Define and describe the principles of Demography, Demographic cycle, Vital statistics
CM9.2	Define, calculate and interpret demographic indices including birth rate, death rate, fertility rates

CM9.3	Enumerate and describe the causes of declining sex-ratio and its social and health Implications
CM9.4	Enumerate and describe the causes and consequences of population explosion and population dynamics of India.
CM9.5	Describe the methods of population control
CM9.6	Describe the National Population Policy
CM9.7	Enumerate the sources of vital statistics including census, SRS, NFHS, NSSO etc

Topic 10: Reproductive maternal and child health

CM10.1	Describe the current status of Reproductive, maternal, newborn and Child Health
CM10.2	Enumerate and describe the methods of screening high-risk groups and common health problems
CM10.3	Describe local customs and practices during pregnancy, child birth, lactation and child feeding practices
CM10.4	Describe the reproductive, maternal, newborn & child health (RMCH); child survival and safe motherhood interventions
CM10.5	Describe Universal Immunization Program; Integrated Management of Neonatal and Childhood Illness (IMNCI) and other existing Programs.
CM10.6	Enumerate and describe various family planning methods, their advantages and shortcomings
CM10.7	Enumerate and describe the basis and principles of the Family Welfare Program including the organization, technical and operational aspects

CM10.8 Describe the physiology, clinical management and principles of adolescent health including ARSH

CM10.9 Describe and discuss gender issues and women empowerment

CM10.10 Able to manage the health care services for reproductive and child care services under supervision

Topic 11: Occupational Health

CM11.1 Enumerate and describe the presenting features of patients with occupational illness including agriculture

CM11.2 Describe the role, benefits and functioning of the employees state insurance scheme

CM11.3 Enumerate and describe specific occupational health hazards, their risk factors and preventive measures

CM11.4 Describe the principles of ergonomics in health preservation

CM11.5 Describe occupational disorders of health professionals and their prevention & management

CM11.6 Able to manage the occupational health services at factory or industry level in a simulated environment

Topic 12: Geriatric services

CM12.1 Define and describe the concept of Geriatric services

CM12.2 Describe health problems of aged population

CM12.3 Describe the prevention of health problems of aged population

CM12.4 Describe National program for elderly

CM12.5 Able to identify the health needs to elderly individuals at the earliest

Topic 13: Disaster Management

CM13.1 Define and describe the concept of Disaster management

CM13.2 Describe disaster management cycle

CM13.3 Describe man-made disasters in the world and in India

CM13.4 Describe the details of the National Disaster management Authority

CM13.5 Able to understand the management of handling a disaster in a simulated environment

Topic 14: Hospital waste management

CM14.1 Define and classify hospital waste

CM14.2 Describe various methods of treatment of hospital waste

CM14.3 Describe laws related to hospital waste management

CM14.4 Able to segregate the various hospital waste

Topic 15: Mental Health

CM15.1 Define and describe the concept of mental Health

CM15.2 Describe warning signals of mental health disorder

CM15.3 Describe National Mental Health program

CM15.4 Able to recognise the mental issues among individuals, families and communities at the earlier stages

Topic 16: Health planning and management

CM16.1	Define and describe the concept of Health planning
CM16.2	Describe planning cycle
CM16.3	Describe Health management techniques
CM16.4	Describe health planning in India and National policies related to health and health planning
CM16.5	Demonstrate understanding of concepts of Health planning in India, various health care economics analysis

Topic 17: Health care of the community

CM17.1	Define and describe the concept of health care to community
CM17.2	Describe community diagnosis
CM17.3	Describe primary health care, its components and principles
CM17.4	Describe National policies related to health and health planning and millennium development goals
CM17.5	Describe healthcare delivery in India
CM17.6	Demonstrate understanding of health system functioning in India

Topic 18: International Health

CM18.1	Define and describe the concept of International health
CM18.2	Describe roles of various international health agencies

CM18.3	Demonstrate understanding role of various international and national agencies in health & disease with prevention of emergence and re-emergence of diseases and prevention of pandemic and handling the Pandemic
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Topic 19: Essential Medicine

CM19.1	Define and describe the concept of Essential Medicine List (EML)
CM19.2	Describe roles of essential medicine in primary health care
CM19.3	Describe counterfeit medicine and its prevention
CM19.4	Demonstrate understanding of mechanism of identifying and calculation of requirements of various medicines and essential medicine at primary health care

Topic 20: Recent advances in Community Medicine

CM20.1	List Important public health events of last five years
CM20.2	Describe various issues during outbreaks and their prevention
CM20.3	Describe any event important to Health of the Community
CM20.4	Demonstrate awareness about laws pertaining to practice of community medicine